Subject: Re: Notes from today's meeting
Date: Thursday, 20 November 2025 at 13:37:38 Greenwich Mean Time
From: Lachlan Black (Student)
To: Leighton Pritchard
Category: BM432, Teaching, UG Projects
Attachments: alligned.fasta

Hi Leighton,

Ncfp is not working and I don't know why.

This is the command I'm using:

ncfp alligned.fasta \
msa_ncfp \
lachlanjblack@gmail.com

I get this output:

```
Process input sequences: 100%|████████████████| 92/92 [00:00<00:00,
1691.33it/s]
Search NT IDs:   0%|                                        | 0/92 [00:08<?, ?
it/s]
Traceback (most recent call last):
  File "/opt/anaconda3/envs/proj313/bin/ncfp", line 10, in <module>
    sys.exit(run_main())
             ~~~~~~~~^^
  File "/opt/anaconda3/envs/proj313/lib/python3.13/site-
packages/ncbi_cds_from_protein/scripts/ncfp.py", line 261, in run_main
    addedrows, countfail = search_nt_ids(
                           ~~~~~~~~~~~~~^
        qrecords, cachepath, args.retries, disabletqdm=args.disabletqdm
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "/opt/anaconda3/envs/proj313/lib/python3.13/site-
packages/ncbi_cds_from_protein/entrez.py", line 237, in search_nt_ids
    result = elink_fetch_with_retries(
        record.id, "protein", "protein_nuccore", retries
    )
ntrez.py", line 318, in elink_fetch_with_retriesncbi_cds_from_protein/e
    raise NCFPMaxretryException(
        "Query ID %s ELink failed\n%s" % (query_id, last_exception())
    )
ncbi_cds_from_protein.entrez.NCFPMaxretryException: Query ID EPD1771162.1
ELink failed
NoneType: None
```

The protein also has an available nucleotide sequence, so I am not sure why it didn't work.
I have attached the input file I'm using.

Thanks,

Lachlan

From: Leighton Pritchard <leighton.pritchard@strath.ac.uk>
Date: Wednesday, 19 November 2025 at 13:47
To: Lachlan Black (Student) <lachlan.black.2022@uni.strath.ac.uk>
Subject: Re: Notes from today's meeting

Hi Lachlan,

If you format the sequence headers appropriately, ncfp will automatically return the nucleotide alignment corresponding to your protein alignment for the sequences it can find, so additional back-threading isn't required.

If you don't want to do that, then the indicated tool (written by my colleague, as it happens), or the t-coffee back translation would both be good enough.

And doing it the wrong way first time is all part of bioinformatics. There's nothing at risk but time - no consumables costs, no explosions...

Cheers,

L.

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If this mail was sent to you or received out of hours, there is no expectation of response until regular working hours.

From: Lachlan Black (Student) <lachlan.black.2022@uni.strath.ac.uk>
Date: Wednesday, 19 November 2025 at 13:15
To: Leighton Pritchard <leighton.pritchard@strath.ac.uk>
Subject: Re: Notes from today's meeting

Thanks Leighton,

For making the nucleotide tree, I am assuming I use my MSA prior to trimming to get the

nucleotide sequences using the tool you linked. Then I thread those onto the alligned sequences with a tool like this: https://github.com/peterjc/pico_galaxy/tree/master/tools/align_back_trans ?

Just want to make sure I am doing it right before I go ahead and don't waste time doing it the wrong way.

Thanks again,

Lachlan

From: Leighton Pritchard <leighton.pritchard@strath.ac.uk>
Date: Wednesday, 19 November 2025 at 08:15
To: Lachlan Black (Student) <lachlan.black.2022@uni.strath.ac.uk>
Subject: Re: Notes from today's meeting

Hi Lachlan,

Thanks for the reminders!

For bootstraps, I mentioned that - for large datasets in particular - the Transfer Bootstrap Expectation, rather then Felsenstein Bootstrap, avoids underestimating the robustness of deep partitions in the tree topology. I don't think you have enough sequences to merit using it over Felsenstein, but out of interest a paper comparing the approaches is here: https://doi.org/10.1093/sysbio/syad052

The more informative thing for you would be to construct a nucleotide tree, as you note. It's almost as simple as making a protein tree, only you'll need to add the steps of (i) identifying the coding sequences and (ii) threading them onto the protein sequence alignment. (NOTE: don't just align the nucleotide sequences!). Happily, this tool should make the job a wee bit simpler: https://ncfp.readthedocs.io/en/stable/ - it may fail to find some corresponding coding sequences, but unless you lose a lot of information that won't be a big problem (and is probably inevitable). Let me know how you get on with it!

Detecting HGT isn't entirely straightforward. There are a number of tools employing quite different approaches and there's no "one true way" to do it. You might try one of these tools:

https://github.com/SysBioChalmers/HGTphyloDetect
https://github.com/GDKO/AvP

but I'm not familiar with their operation, so I'll look forward to seeing how intuitive you find them 😉

Cheers,

L.

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If this mail was sent to you or received out of hours, there is no expectation of response until regular working hours.

From: Lachlan Black (Student) <lachlan.black.2022@uni.strath.ac.uk>
Date: Tuesday, 18 November 2025 at 14:23
To: Leighton Pritchard <leighton.pritchard@strath.ac.uk>
Subject: Re: Notes from today's meeting

Thanks Leighton,

I forgot to take any notes yesterday and have forgot some of the stuff we discussed that I could work on.

Could you please remind me what you suggest I do regarding the bootstrap values? I think you suggested a specific analysis I should do but I completely forgot.

Also, if I remember correctly, you suggest that I backtrace the protein sequences and building a phylogenetic tree from the DNA sequences? Is that as simple as just repeating the same steps but with the DNA sequences?

And finally, you suggested an analysis I can perform regarding horizontal gene transfer between E. coli and Shigella, but I forget what it was.

I promise to take notes next time 😅, and if you need anything from me to remember what we discussed yesterday I'll send it over.

Lachlan

From: Leighton Pritchard <leighton.pritchard@strath.ac.uk>
Date: Monday, 17 November 2025 at 16:47
To: Iona Milligan (Student) <iona.milligan.2022@uni.strath.ac.uk>, Ash Eana (Student) <ash.eana.2021@uni.strath.ac.uk>, Lachlan Black (Student) <lachlan.black.2022@uni.strath.ac.uk>, Liam Taggart (Student) <liam.taggart.2022@uni.strath.ac.uk>, Jingyang Tong (Student) <jingyang.tong.2022@uni.strath.ac.uk>

Subject: Notes from today's meeting

Hi all,

Many thanks to everyone who presented today - I'm confident that you've all got interesting proteins about which a *lot* can be said and proposed in terms of evolution and function. We've got two working weeks left, formally-speaking, but as I mentioned in the meeting computational projects can be stretched a little bit into time that others can't get into the lab, so you're not quite as time-constricted as some of your classmates.

In the meeting we talked a little bit about ETE3 for phylogenetic tree drawing - mostly concerning its not very intuitive syntax if you haven't come across that style of Python - and I said I'd attach an example Jupyter Notebook so you can see how I've structured some more complex tree labelling in the past.

Jingyang had had difficulties with the Colab Alphafold server in that it was not returning predictions for heptamers of his protein. Lachlan had used the Alphafold Server service with larger multimers of his protein, so suggested it as a good choice: https://alphafoldserver.com/welcome

Next week's meeting is on Tuesday 25th at 1:30 in HW324.

Also, there's a seminar tomorrow at 2pm in CW404b for anyone who's interested: Fabrice Graf from LSTM is presenting "An old drug against superbugs? Molecular mechanisms of re-emerging chloramphenicol susceptibility in multidrug resistant Enterobacterales"

I'll see you all next week on Tuesday, but if you need anything before then please let me know by email, or we can arrange a meeting.

Cheers,

L.

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